The Conformational Instability of Folded Full-Length Staphylococcal Protein A Revealed by Simulation and Experiment

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Background and Motivation

Multi-domain proteins

Most research studies focus on single protein domains.

However, 75% of the eukaryotic proteome comprises multidomain proteins.

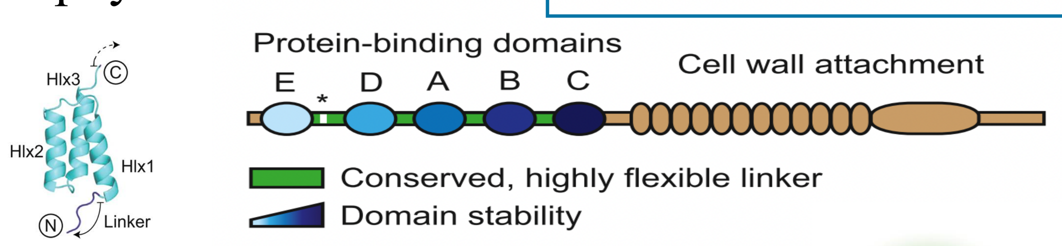
Han, Jung-Hoon, et al. *Nature reviews Molecular cell biology* 8, no. 4 (2007): 319-330.

Staphylococcal protein A (SpA)

>> Major surface protein present in almost all strains of Staphylococcus aureus

N-terminal half, consists of five protein-binding domains (**E-D-A-B-C**)

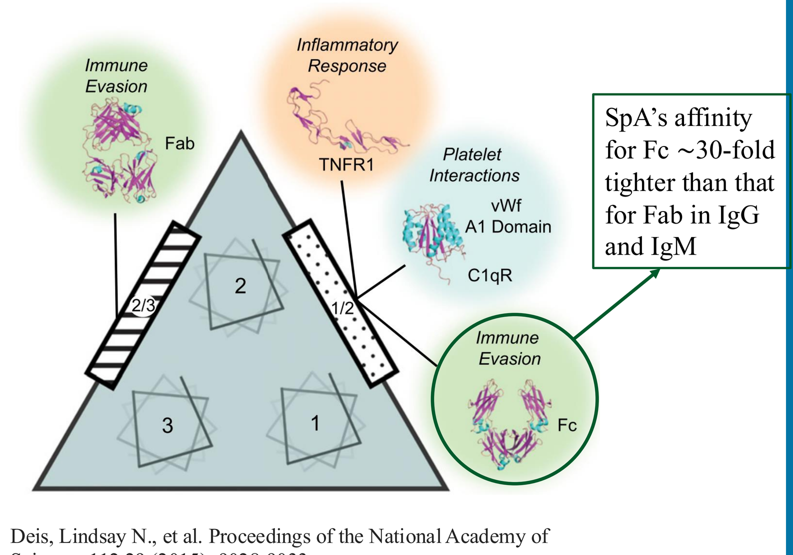
C-terminal half, **region X**, cell-wall anchoring region



Deis, Lindsay N., et al. *Structure* 22, no. 10 (2014): 1467-1477.

SpA as a virulence factor

>> SpA has a wide range of functions that require binding to many target proteins in the host.



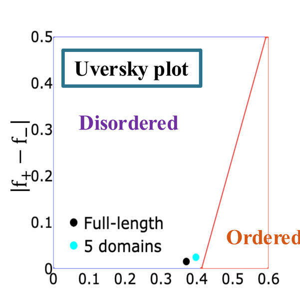
SpA's affinity for Fc ~30-fold tighter than that for Fab in IgG and IgM

Deis, Lindsay N., et al. Proceedings of the National Academy of Sciences 112.29 (2015): 9028-9033.

Motivation

>> Clinical trials targeting SpA: largely ineffective to date

>> Contrasting views on full-length SpA characteristics.



Howden, Benjamin P., et al., *Nature Reviews Microbiology* 21, no. 6 (2023): 380-395.

The whole protein cannot be investigated with crystallography

Also computationally demanding

Employing enhanced sampling molecular dynamics simulations to **atomistically** model the full-length SpA for the first time

Simulation Methods

>> Force field: **Charmm36** for protein >>> Modified TIP3P for water

>> Sampling method:

>>> **PTWTE-WTM** (Parallel-Tempering in the Well-Tempered Ensemble combined with Well-Tempered Metadynamics)

>>> **MM-OPES** (Multithermal-Multiumbrella On-the-fly Probability Enhanced Sampling)

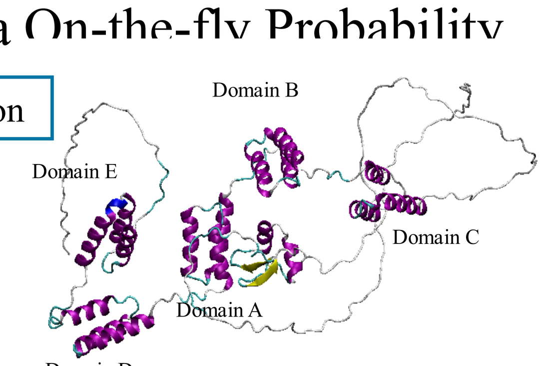
>> Order parameter

Reference: AlphaFold prediction

Q — Used for biasing, based on the distance between each atomic pair considered in contact

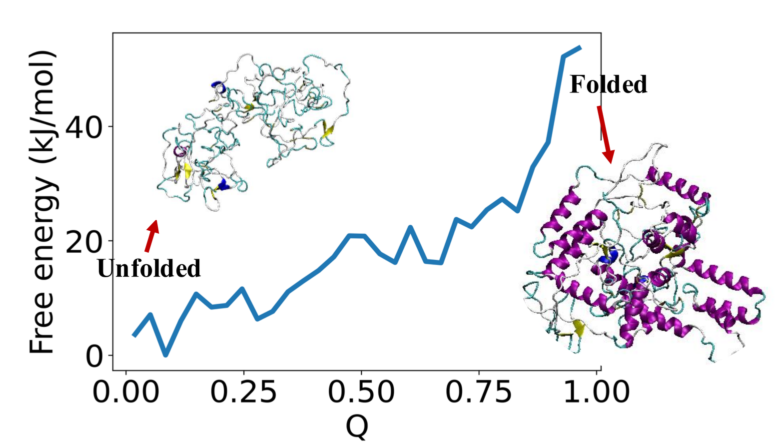
$$Q = \frac{1}{N_{nb_{nem}}} \sum_{(i,j)} \frac{1}{1+\exp(\gamma(r_{ij} - \lambda r_{ij}^{0}))}$$

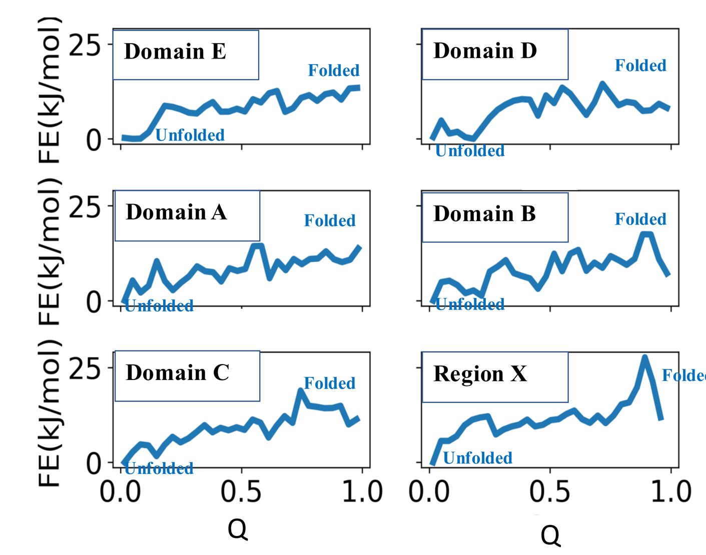
Q = 1
- Totally **Folded** domains



Results and Discussions

Free energy calculated from **full-length** protein A simulation

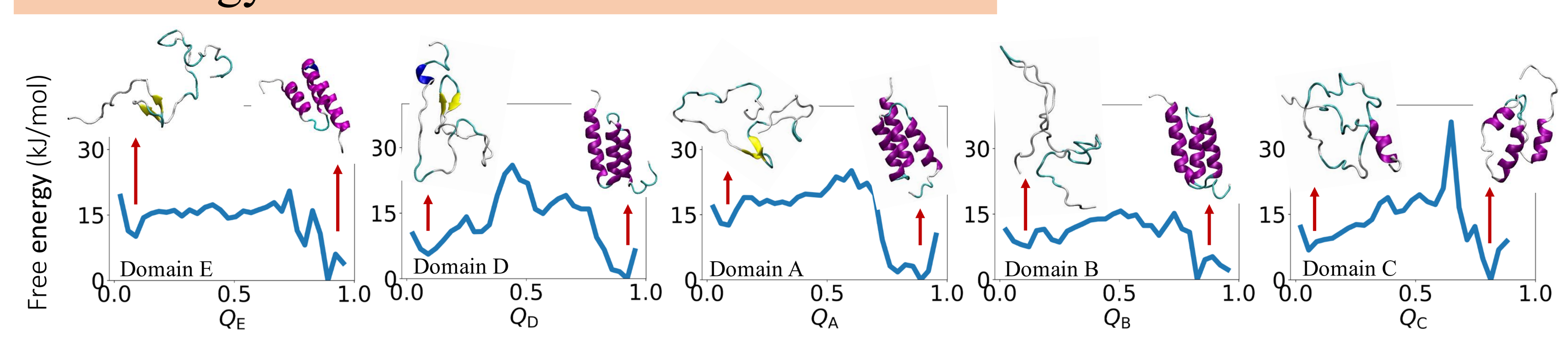




>> The **folded** configuration of full-length SpA is **unstable** at room temperature.

>> None of the domains can fold in the full-length assembly.

Free energy of individual domains in isolation

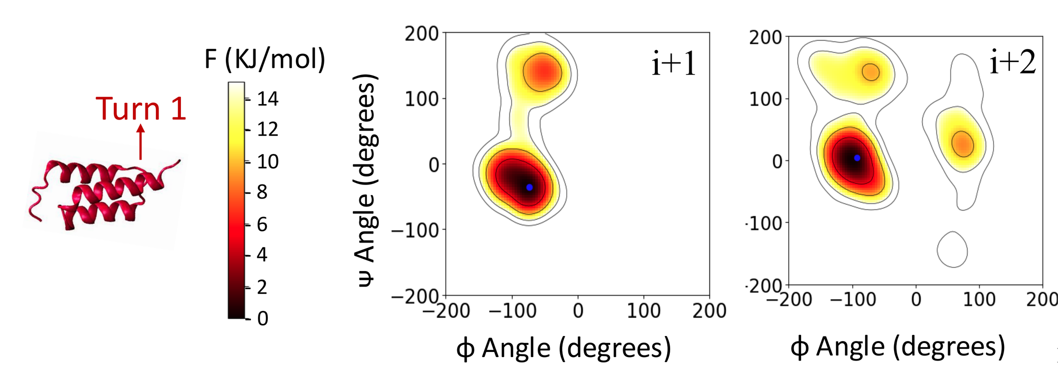


>> **Folded** individual domains are stable **in isolation**.

>> The instability of full-length SpA is **not an artifact of the force field.**

Ramachandran plot for the first turn of the domain E

>> Domain E's turn 1 in the **isolated** domain simulation

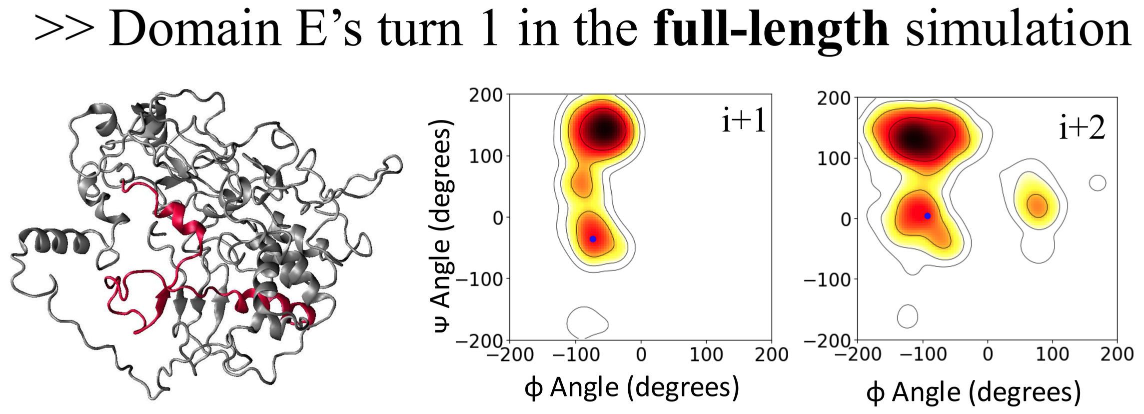


Reported for helix-turn-helix

Turn Type	Dihedral angles (°)			
	ϕi+1	ψi+1	ϕi+2	ψi+2
I	-60	-30	-90	0

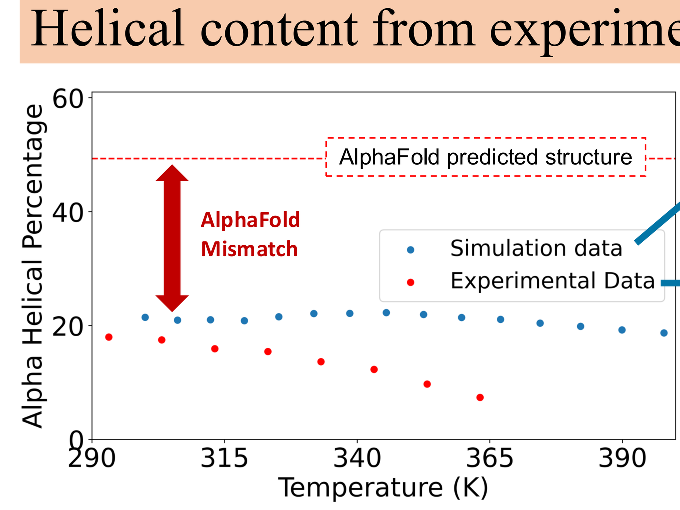
Hutchinson, E. Gail. *Protein Science* 3.12 (1994): 2207-2216.

>> Domain E's turn 1 in the **full-length** simulation



>> ϕ and ψ angles analysis does **not** show preference of turn formation.

Helical content from experiment and simulation

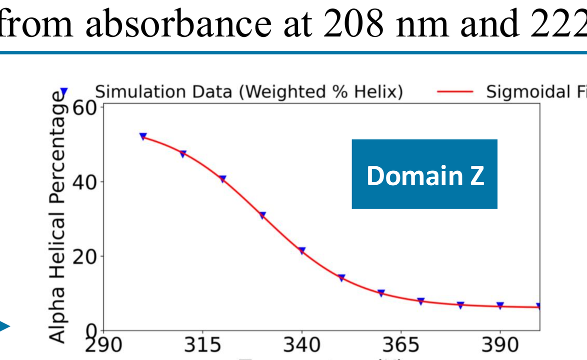


Using the **DSSP** (Dictionary of Secondary Structures in Proteins) algorithm for 20 replicas

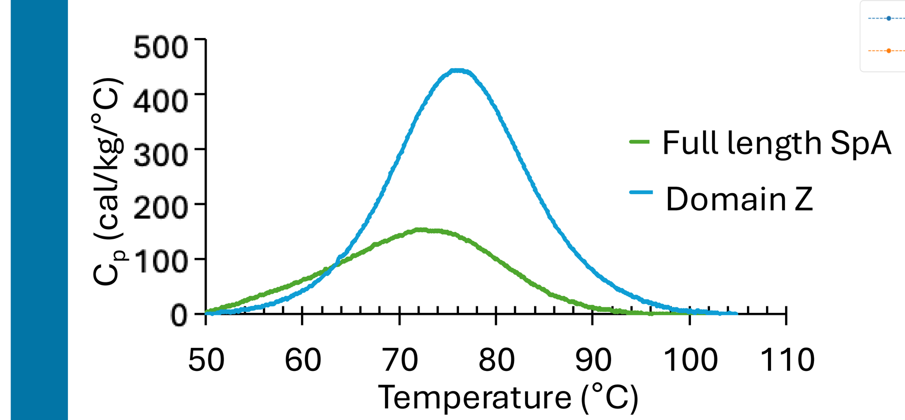
Circular Dichroism (CD) spectroscopy Calculated from absorbance at 208 nm and 222 nm

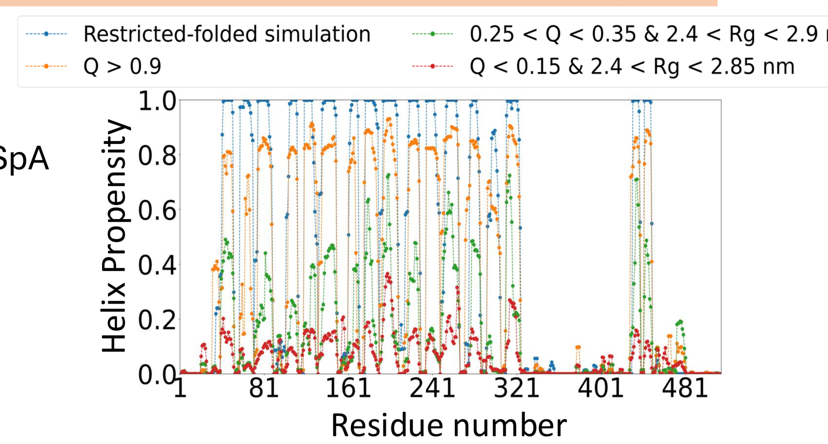
>> Full-length protein A shows **low helicity with no transition**

2 states folding



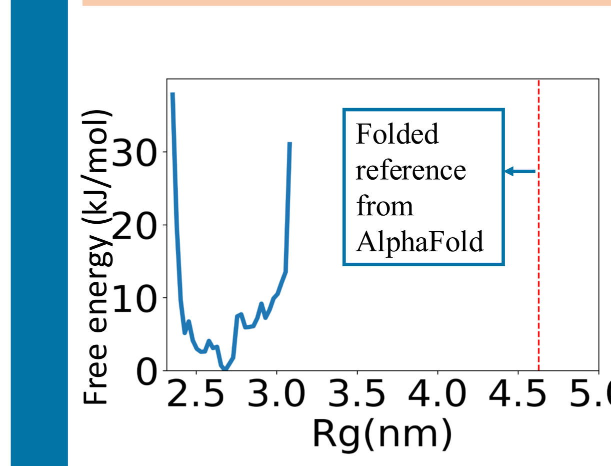
DSC thermogram and helix propensity analysis of full-length SpA

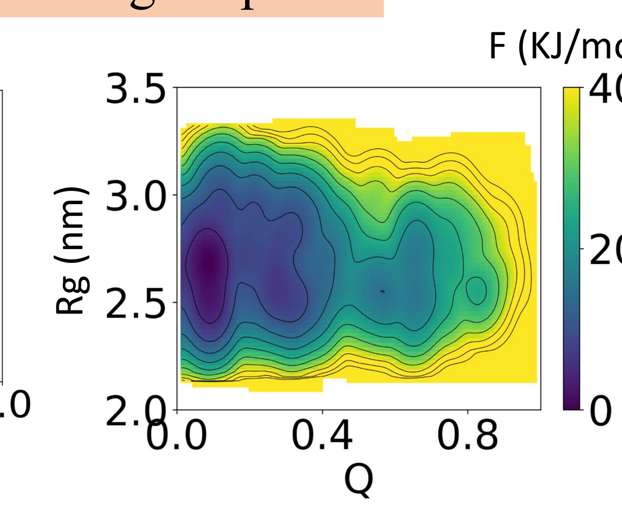




>> **Broad thermal transition** in SpA → Partial helical formation

Radius of gyration of full-length SpA

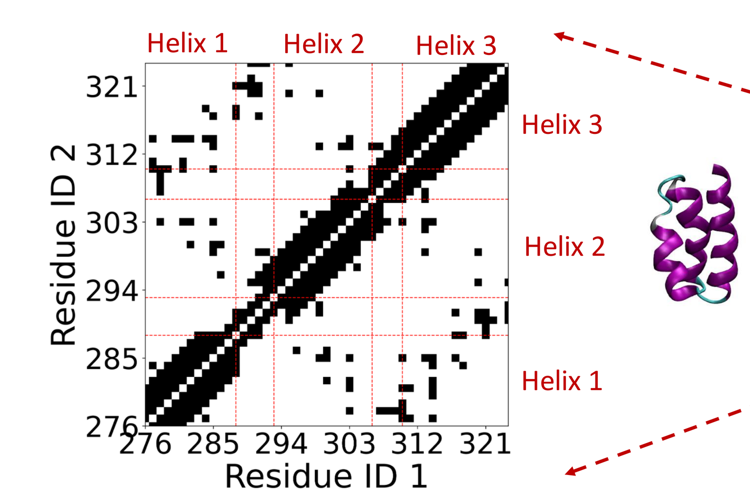




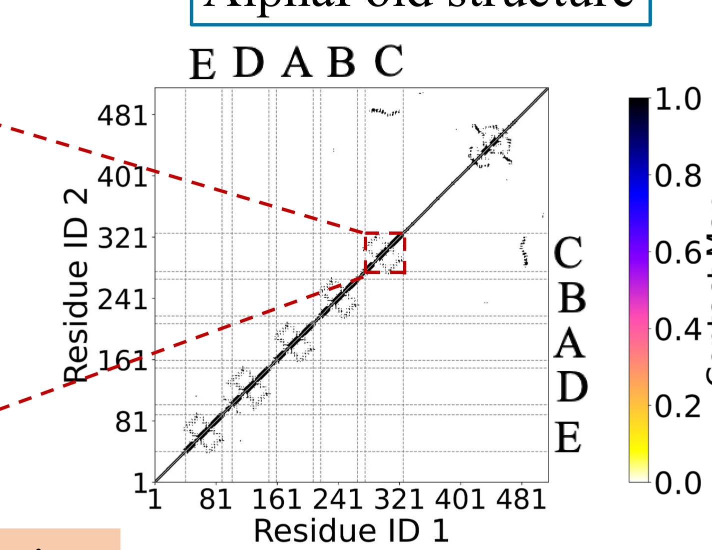
>> R_g has a **narrow** range of no more than 3.1 nm

Compact structure

Contact map of full-length SpA- AlphaFold

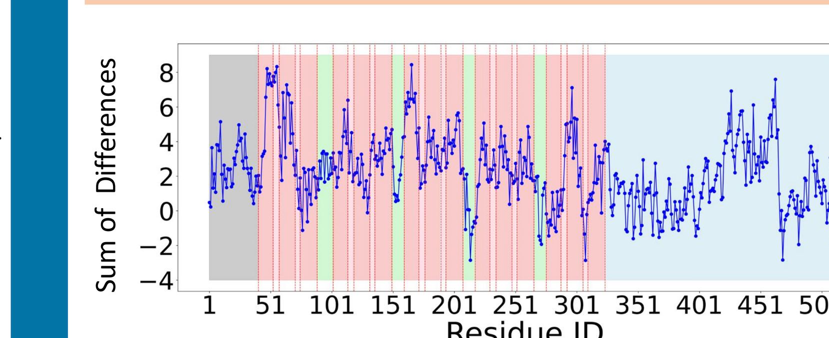


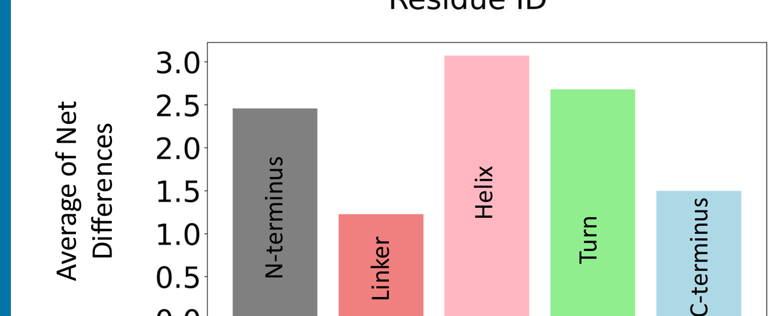
AlphaFold structure



Contact map of full-length SpA- Simulation

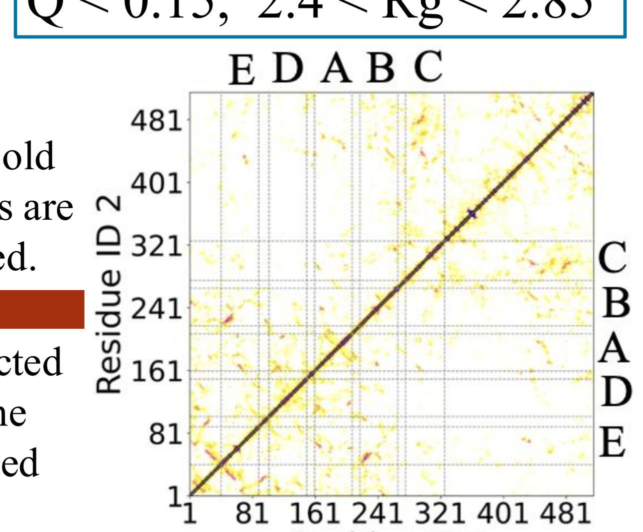
$Q < 0.15, \ 2.4 < Rg < 2.85$





AlphaFold contacts are excluded.

Subtracted from the restricted folded simulation



>> The **helix** region contributes most to the unpredicted contacts.

>> 75 % of amino acids with high propensities are **Hydrophobic**.

Conclusions

>> Individual domains of protein A are stable in isolation. However, when they are placed in the full-length SpA structure, the full-length protein becomes **unstable**.

>> **Unpredicted contacts** may significantly contribute to the compactness of the structure. Steric hindrance created by **compactness** prevents folding.

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